

### Wide format (raw)

*rows = cell lines, columns = genes*

| profile_id | TSPAN6 | TNMD | DPM1 | ... |
|------------|--------|------|------|-----|
| PR-AdBjpG  | 4.33   | 0    | 7.36 | ... |
| PR-I2AzwG  | 4.56   | 0.58 | 7.11 | ... |
|            |        | ⋮    |      |     |

**reshape + map**

*profile\_id → ach\_id, 1 row → 3 rows*

### Long format (query-ready)

*each row = one cell line × gene combination*

| profile_id | ach_id     | gene   | value |
|------------|------------|--------|-------|
| PR-AdBjpG  | ACH-001113 | TSPAN6 | 4.33  |
| PR-AdBjpG  | ACH-001113 | TNMD   | 0     |
| PR-AdBjpG  | ACH-001113 | DPM1   | 7.36  |
| PR-I2AzwG  | ACH-001289 | TSPAN6 | 4.56  |
| PR-I2AzwG  | ACH-001289 | TNMD   | 0.58  |
| PR-I2AzwG  | ACH-001289 | DPM1   | 7.11  |
|            | ⋮          |        |       |